

COMP-SCI-5565

Intro to Statistical Learning

Topic 3 – Classification

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Lecture Objectives

- Compare and contrast classification with linear regression.
- Perform classification using logistic regression.
- Perform classification using linear discriminant analysis (LDA).
- Perform classification using naive Bayes.
- Perform classification using K-nearest Neighbors.
- Identify the strengths and weaknesses of the various classification models.

Outline

3.1 Overview of Classification

3.2 Logistic Regression

3.3 Linear Discriminant Analysis

3.4 Naïve Bayes

4.5 K-nearest Neighbors

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3.1 Overview of Classification

3.2 Logistic Regression

3.3 Linear Discriminant Analysis

3.4 Naïve Bayes

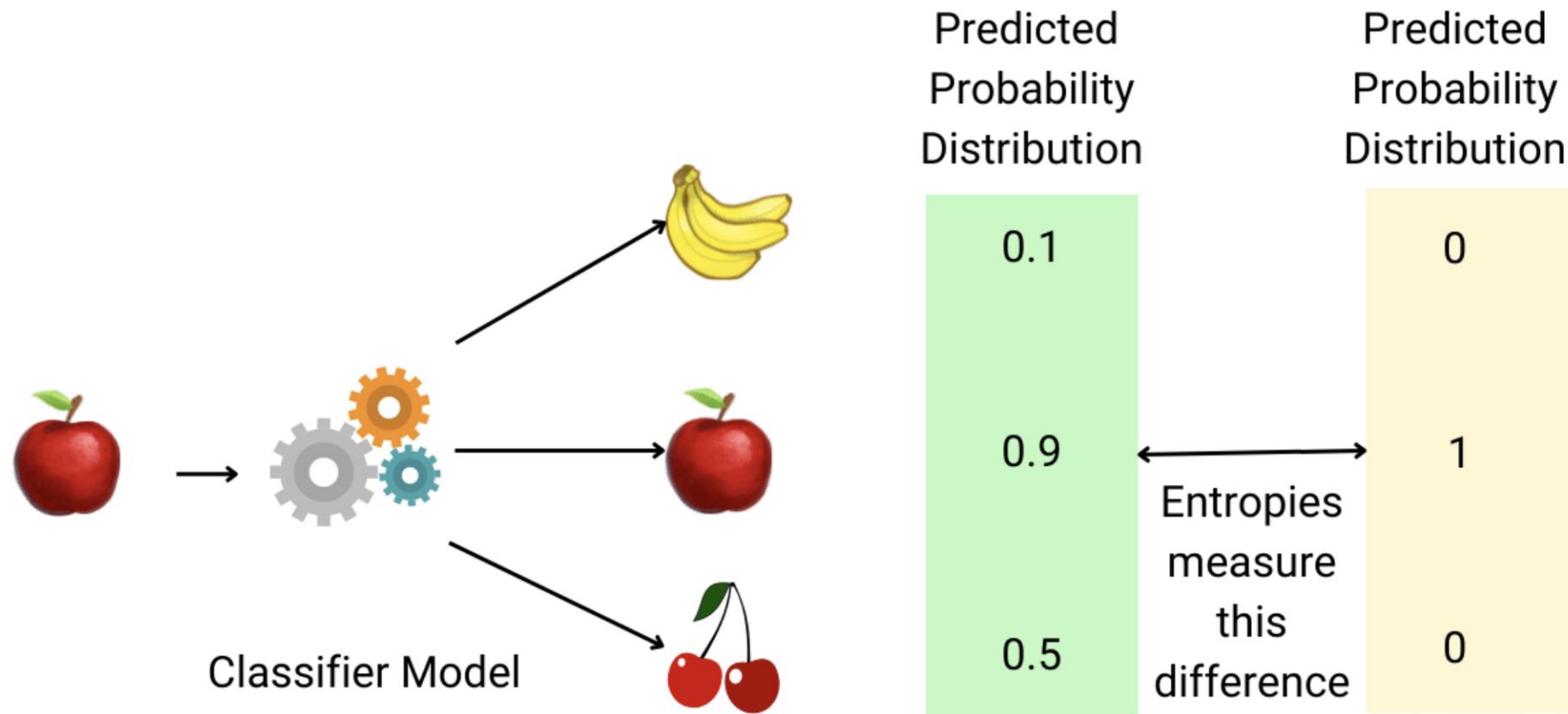
4.5 K-nearest Neighbors

3.1 Overview of Classification

- Classification methods make inferences and/or predict qualitative (categorical) response variables.
- E.g., Predicting fraudulent activity (Yes/No)
- Qualitative variables take values in an unordered set $\mathcal{C}$, such as:
 - $\text{eye color} \in \{\text{blue}, \text{brown}, \text{green}\}$
 - $\text{email} \in \{\text{spam}, \text{ham}\}$
- Given a feature vector X and a qualitative response Y taking values in the set $\mathcal{C}$.
- The classification task is to build a function $C(X)$ that takes as input the feature vector X and predicts its value for Y i.e., $C(X) = Y \in \mathcal{C}$.

3.1 Overview of Classification

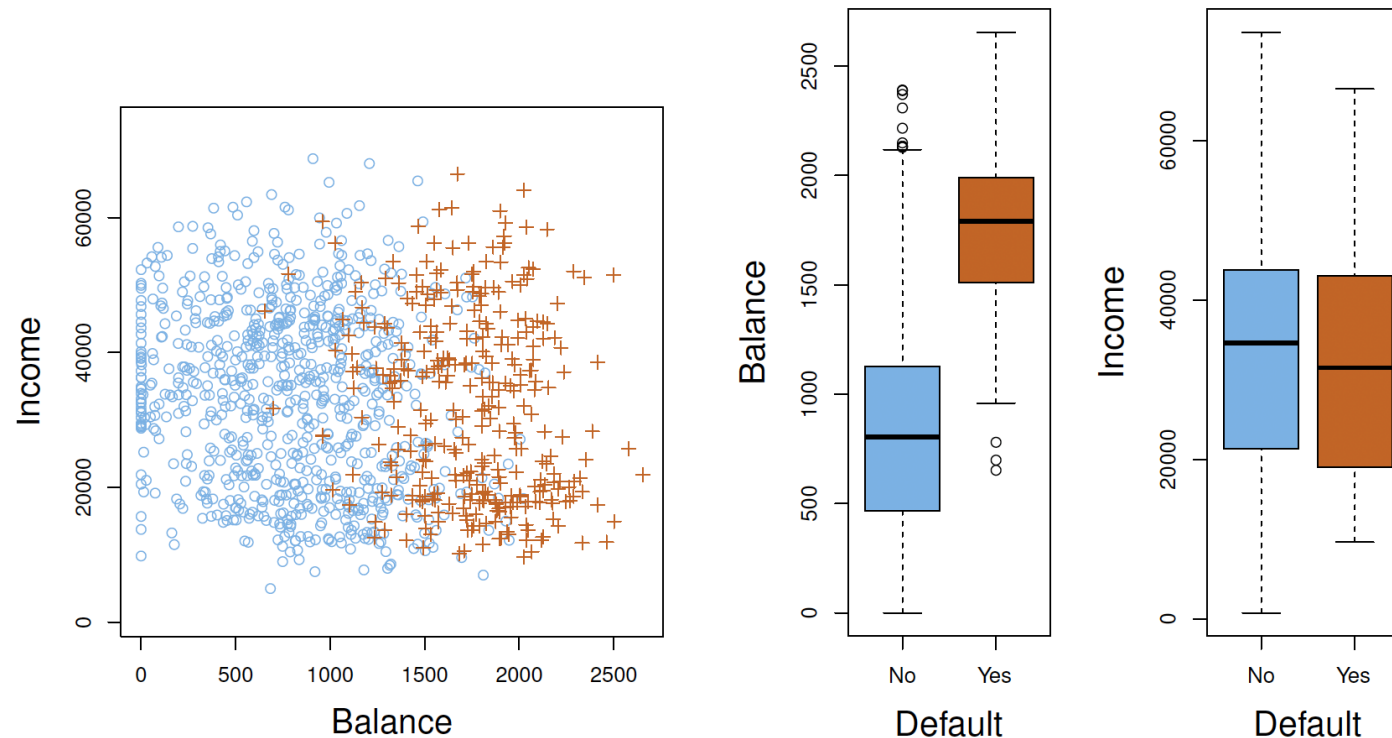
Often, we are more interested in estimating the probabilities that X belongs to each category in $\mathcal{C}$.



3.1 Overview of Classification

Example

The distribution of balance and income is split by the binary default variable, respectively. Defaulters are represented as the orange plus sign; non-defaulters are represented as the blue circle.



3.1 Overview of Classification

Why NOT Linear Regression?

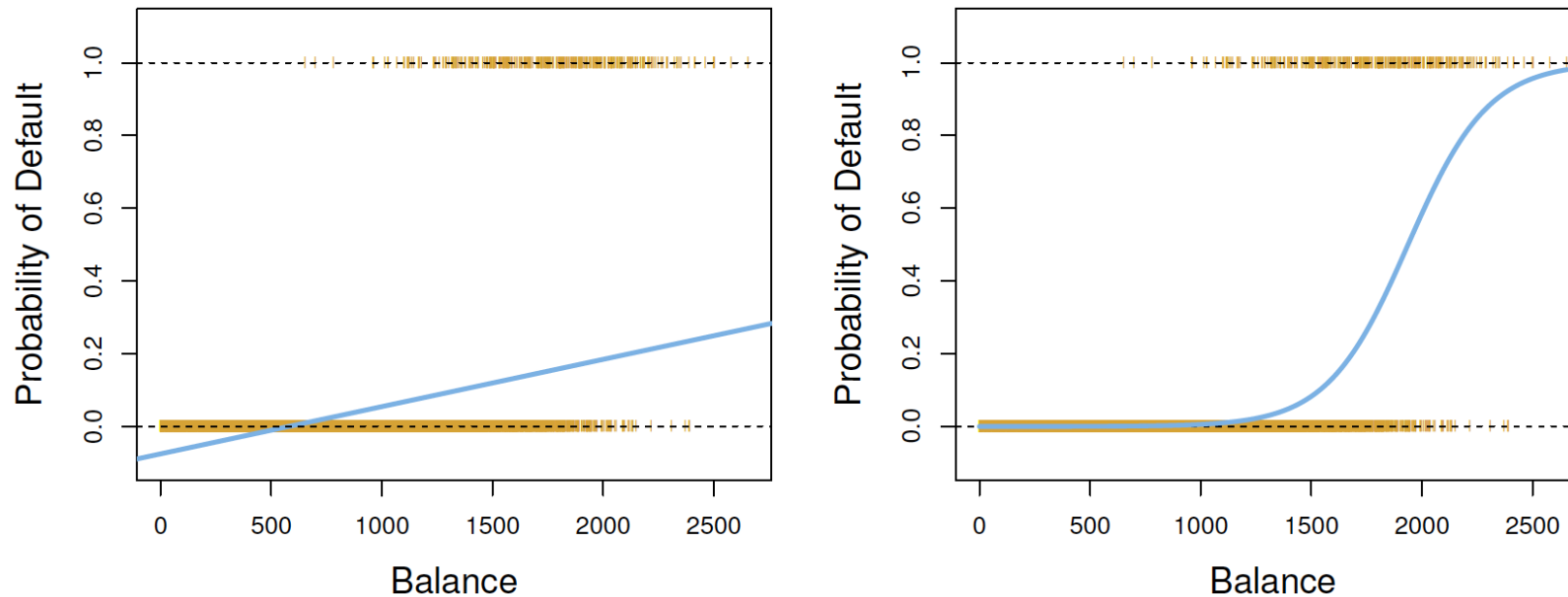
- A regression method cannot always convert a qualitative response variable into a quantitative response ready for linear regression.
- Suppose for the **Default** classification task, we code

$$Y = \begin{cases} 0 & \text{if No} \\ 1 & \text{if Yes} \end{cases}$$

- Can we simply perform a linear regression of Y on X and classify as **Yes** if $\hat{Y} > 0.5$?
- Linear regression might produce probabilities less than zero or bigger than one.
- *Logistic regression* is more appropriate.

3.1 Overview of Classification

Why NOT Linear Regression? – continued



Classification using the Default data. Left: Estimated probability of default using linear regression.

Some estimated probabilities are negative! The orange ticks indicate the 0/1 values coded for default(No or Yes). Right: Predicted probabilities of default using logistic regression. All probabilities lie between 0 and 1.

3.1 Overview of Classification

Why NOT Linear Regression? – continued

- Now, we have a response variable with three possible values. A patient presents at the emergency room, and we must classify them according to their symptoms.

$$Y = \begin{cases} 1 & \text{if stroke} \\ 2 & \text{if drug overdose} \\ 3 & \text{if epileptic seizure.} \end{cases}$$

- This coding suggests an ordering and implies that the difference between **stroke** and **drug overdose** is the same as between **drug overdose** and **epileptic seizure**.
- Linear regression is not appropriate here.
- *Multiclass Logistic Regression* or *Discriminant Analysis* are more appropriate.

Outline

3.1 Overview of Classification

3.2 Logistic Regression

3.3 Linear Discriminant Analysis

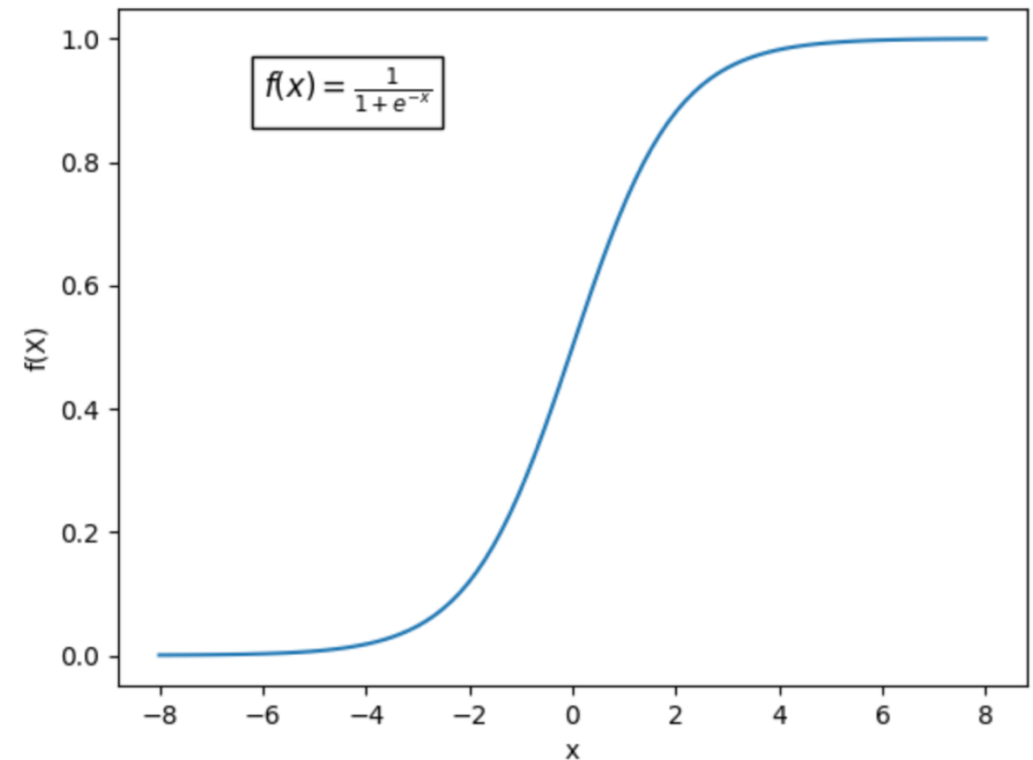
3.4 Naïve Bayes

4.5 K-nearest Neighbors

3.2 Logistic Regression

The Logistic Regression Model

- Logistic regression models the probability that X belongs to a particular category, Y .
$$p(X) = Y$$
- Y is binary – 0/1.
- Logistic regression uses a *sigmoid function* to map X and their probabilities.
- The formula of the sigmoid function is
$$f(x) = \frac{1}{1+e^{-x}},$$
where e is Euler's number ≈ 2.71 .
- The function that takes any real number and maps it to a probability between 1 and 0.



3.2 Logistic Regression

The Logistic Regression Model– continued

- Linear regression uses the form

$$p(X) = \beta_0 + \beta_1 X$$

- Logistic regression uses the form

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

3.2 Logistic Regression

Estimating Coefficients

- To estimate the regression coefficient, we use *maximum likelihood* (ME).
- Likelihood Function

$$\ell(\beta_0, \beta) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i))$$

- The aim is to find β_0 and β_1 to maximize the likelihood ℓ of the observed data x_i .
- The Least square method is the special case of maximum likelihood function.

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.6513	0.3612	-29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

3.2 Logistic Regression

Making Predictions

- What is our estimated probability of **default** for someone with a balance of \$1000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 1000}}{1 + e^{-10.6513 + 0.0055 \times 1000}} = 0.006$$

- With a **balance** of \$2000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 2000}}{1 + e^{-10.6513 + 0.0055 \times 2000}} = 0.586$$

3.2 Logistic Regression

Multiple Logistic Regression

- We add more coefficients.

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}$$

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.8690	0.4923	-22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income	0.0030	0.0082	0.37	0.7115
student [Yes]	-0.6468	0.2362	-2.74	0.0062

3.2 Logistic Regression

Multinomial Logistic Regression

- In this situation, we have more than two classes.
- A multinomial logistic regression can have the form

$$\Pr(Y = k | X) = \frac{e^{\beta_{0k} + \beta_{1k}X_1 + \dots + \beta_{pk}X_p}}{\sum_{\ell=1}^K e^{\beta_{0\ell} + \beta_{1\ell}X_1 + \dots + \beta_{p\ell}X_p}}$$

- We select a single class to serve as the baseline for each class.
- The interpretation of the coefficients in a multinomial logistic regression model must be done with care, since it is tied to the choice of baseline.
- Other classification models – KNN, Decision Trees, SVM – are preferred in such situations.

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3.3 Linear Discriminant Analysis

Why Logistic Regression is not ideal?

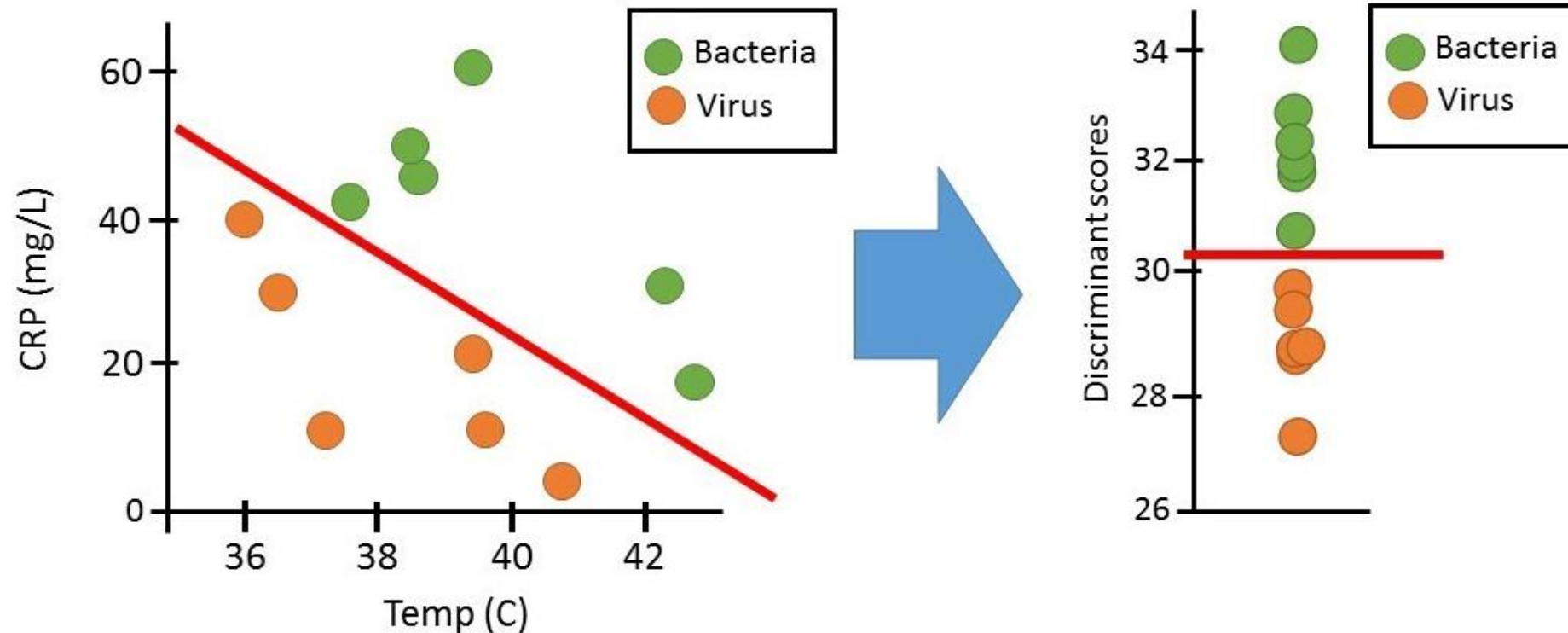
- When there is a substantial separation between the two classes, the parameter estimates for the logistic regression model are surprisingly unstable.
- It is challenging to extend to more than two response classes.
- If the distribution of the predictors X is approximately normal in each class and the sample size is small, then the generative modeling may be more accurate than logistic regression.
- Generative modeling can be naturally extended to the case of more than two response classes.

3.3 Linear Discriminant Analysis

- Linear Discriminant Analysis (LDA) is a parametric approach.
- LDA models the distribution of X in each class separately and then uses the *Bayes theorem* to flip things around and obtain $\Pr(Y|X)$.
- When we use normal (Gaussian) distributions for each class, this leads to linear or quadratic discriminant analysis.
- This approach is quite general, and other distributions can also be used.
- We will focus on normal distribution.

3.3 Linear Discriminant Analysis

- The goal of LDA is to map the samples to some discriminant scores to maximize the distance between classes.
- Then use the mean of the discriminant scores to separate the classes.



3.3 Linear Discriminant Analysis

Bayes' Theorem for Classification

- Bayes' Theorem states that

$$\Pr(Y = k \mid X = x) = \frac{\Pr(X=x|Y=k) \cdot \Pr(Y=k)}{\Pr(X=x)}$$

- $\Pr(Y = k \mid X = x)$ is the probability of class K given X .
- $\Pr(Y = k)$ is the marginal/prior/independent probability for class K .
- $\Pr(X = x)$ is the marginal/prior/independent probability for X .
- $\Pr(X = x \mid Y = k)$ is the probability of X given that K is true (posterior probability).

3.3 Linear Discriminant Analysis

Bayes' Theorem for Classification – continued

- Imagine 100 people at a party, and you tally how many wear pink or not, and if a man or not, and get these numbers:

	Pink	notPink	
Man	5	35	40
notMan	20	40	60
	25	75	100

$$P(\text{Pink} \mid \text{Man}) = \frac{P(\text{Man} \mid \text{Pink}) \cdot P(\text{Pink})}{P(\text{Man})}$$

- The probability of being a man is $P(\text{Man}) = \frac{40}{100} = 0.4$
- The probability of anyone wearing pink is $P(\text{Pink}) = \frac{25}{100} = 0.25$
- The probability that a man wears pink is $P(\text{Man} \mid \text{Pink}) = \frac{5}{25} = 0.2$
- $P(\text{Pink} \mid \text{Man}) = \frac{0.2 \times 0.25}{0.4} = 0.07$

3.3 Linear Discriminant Analysis

Bayes' Theorem for Classification – continued

- Imagine 100 people at a party, and you tally how many wear pink or not, and if a man or not, and get these numbers:

	Pink	notPink	
Man	5	35	40
notMan	20	40	60
	25	75	100

$$P(\text{Pink} \mid \text{Not Man}) = \frac{P(\text{Not Man} \mid \text{Pink}) \cdot P(\text{Pink})}{P(\text{Not Man})}$$

- The probability of not a man is $P(\text{Not Man}) = \frac{60}{100} = 0.6$
- The probability of anyone wearing pink is $P(\text{Pink}) = \frac{25}{100} = 0.25$
- The probability that not a man wears pink is $P(\text{Not Man} \mid \text{Pink}) = \frac{20}{25} = 0.8$
- $P(\text{Pink} \mid \text{Not Man}) = \frac{0.8 \times 0.25}{0.6} = 0.33$

3.3 Linear Discriminant Analysis

- We rewrite Bayes Theorem as

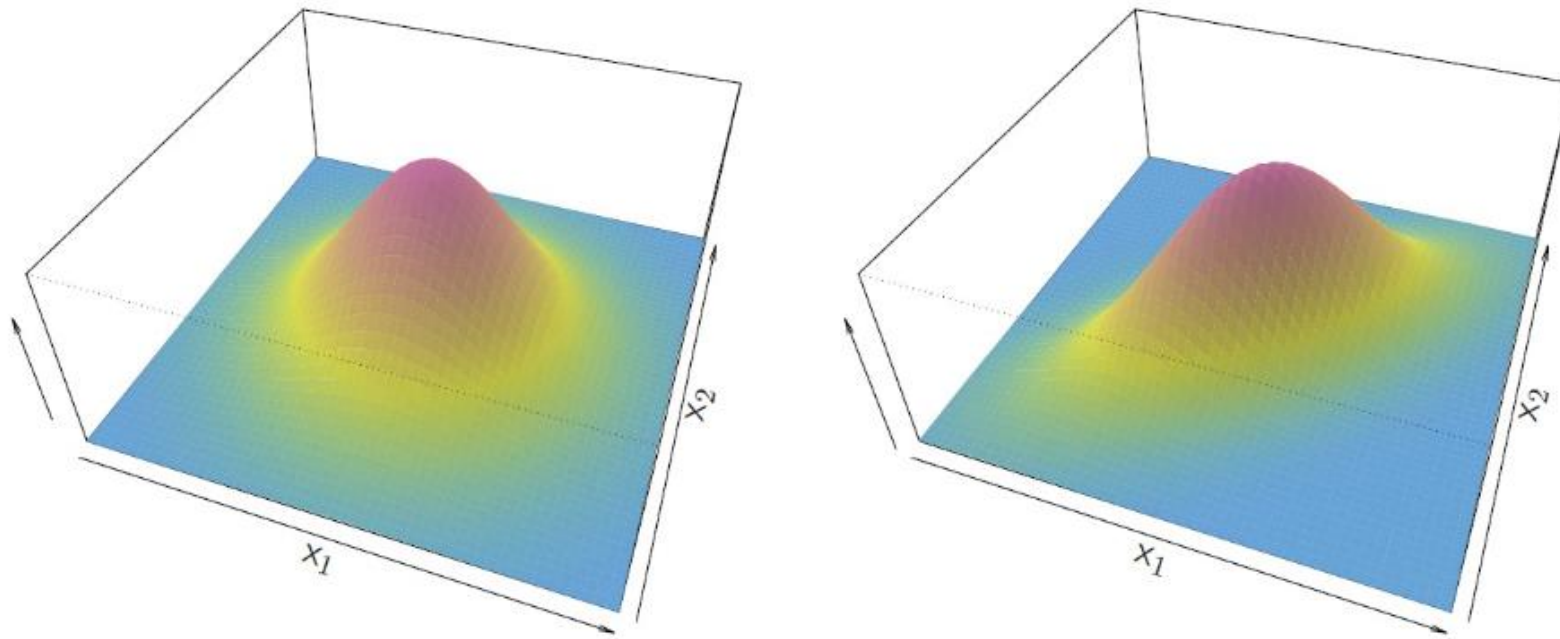
$$p_k = \frac{\pi_k f_k(x)}{\sum_{l=1}^k \pi_l f_l(x)},$$

- $p_k = \Pr(Y = k \mid X = x)$ is the posterior probability that an observation posterior $X=x$ belongs to the k^{th} class.
- $\pi_k = \Pr(Y = k)$ is the overall or prior probability that a randomly chosen observation comes from the k^{th} class.
- $f_k(x) = \Pr(X = x \mid Y = k)$ is the density function of X density for an observation that comes from the k^{th} class; **requires some underlying assumption to estimate.**

3.3 Linear Discriminant Analysis

Assumptions:

- LDA assumes that $f_k(x)$ is **normal** or **Gaussian** with a class specific mean and, a shared variance term across all K classes [$\sigma_1^2 = \dots = \sigma_K^2$].
- Each individual predictor follows a one-dimensional normal distribution, with predictors having some correlation



3.3 Linear Discriminant Analysis

- The normal density function takes the form:

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2} (x - \mu_k)^2\right)$$

where σ is the standard deviation and μ is the mean

- Then, the posterior probability (probability that the observation belongs to the k th class, given the predictor value for that observation) is

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} (x - \mu_k)^2\right)}{\sum_{l=1}^k \pi_l \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} (x - \mu_l)^2\right)}$$

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3.4 Naïve Bayes

- Estimating a p-dimensional density function is challenging; Naïve Bayes make a different assumption than LDA.
- An alternative to LDA that does not assume normally distributed predictors

$$f_k(x) = f_{k1}(x_1) \times f_{k2}(x_2) \times \cdots \times f_{kp}(x_p)$$

where f_{kj} is the density function of the j^{th} predictor among observations in the k th class.

$$\Pr(Y = k \mid X = x) = \frac{\Pr(x_1|k) \cdot \Pr(x_2|k) \cdots \Pr(x_p|k) \cdot \Pr(Y=k)}{\Pr(X=x)}$$

- *Within the k th class, the p predictors are independent.*

3.4 Naïve Bayes

Example

- Suppose we are building a classifier that says whether a text is about sports or not. Our training data has 5 sentences:

Text	Tag
"A great game"	Sports
"The election was over"	Not sports
"Very clean match"	Sports
"A clean but forgettable game"	Sports
"It was a close election"	Not sports

- Now, which tag does the sentence *"A very close game"* belong to?

3.4 Naïve Bayes

Example – continued

- Using word frequencies, we obtain the following features:

	a	great	game	the	election	was	over	very	clean	match	but	forgettable	it	close
S	$\frac{2}{11}$	$\frac{1}{11}$	$\frac{2}{11}$	0	0	0	0	$\frac{1}{11}$	$\frac{2}{11}$	$\frac{1}{11}$	$\frac{1}{11}$	$\frac{1}{11}$	0	0
NS	$\frac{1}{9}$	0	0	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{2}{9}$	$\frac{1}{9}$	0	0	0	0	0	$\frac{1}{9}$	$\frac{1}{9}$

- Total words = 20
- $\Pr(S) = \frac{11}{20}$ and $\Pr(NS) = \frac{9}{20}$
- S = Sports and NS = Not Sports

3.4 Naïve Bayes

Example – continued

- We want to find the $\Pr(\text{Sports} \mid \text{a very close game})$ and $\Pr(\text{Not Sports} \mid \text{a very close game})$ and select the class with the largest probability.

$$\Pr(S \mid \text{a very close game}) = \frac{\Pr(a \mid S) \times \Pr(\text{very} \mid S) \times \Pr(\text{close} \mid S) \times \Pr(\text{game} \mid S) \times \Pr(S)}{\Pr(a) \times \Pr(\text{very}) \times \Pr(\text{close}) \times \Pr(\text{game})}$$

$$\Pr(NS \mid \text{a very close game}) = \frac{\Pr(a \mid NS) \times \Pr(\text{very} \mid NS) \times \Pr(\text{close} \mid NS) \times \Pr(\text{game} \mid NS) \times \Pr(NS)}{\Pr(a) \times \Pr(\text{very}) \times \Pr(\text{close}) \times \Pr(\text{game})}$$

- Since the denominators are the same, we can ignore them and compare the numerators.

3.4 Naïve Bayes

Example – continued

$$\begin{aligned}\Pr(S \mid \text{a very close game}) &= \Pr(a \mid S) \times \Pr(\text{very} \mid S) \times \Pr(\text{close} \mid S) \times \Pr(\text{game} \mid S) \times \Pr(S) \\ &= \frac{2}{11} \times \frac{1}{11} \times 0 \times \frac{2}{11} \times \frac{11}{20} = 0\end{aligned}$$

$$\begin{aligned}\Pr(NS \mid \text{a very close game}) &= \Pr(a \mid NS) \times \Pr(\text{very} \mid NS) \times \Pr(\text{close} \mid NS) \times \Pr(\text{game} \mid NS) \times \Pr(NS) \\ &= \frac{1}{9} \times 0 \times \frac{1}{9} \times 0 \times \frac{9}{20} = 0\end{aligned}$$

- However, we run into a problem here: “close” doesn’t appear in any Sports text, and “game” doesn’t appear in any Not Sports text!
- This is due to the naïve assumptions of predictor independence.

3.4 Naïve Bayes

Example – continued

- Solution - **Laplace smoothing** – we add 1 to every count, so it's never zero.

$$P_{LAP,k}(x) = \frac{c(x) + k}{N + k|X|}$$

where c is the count, k is a non-zero value, N is the total sample size of interest, and $|X|$ is the cardinality of X (how many different values are possible for X).

- Since the number of possible words is 14, applying smoothing we get that

$$\Pr(\text{game} \mid S) = \frac{2 + 1}{11 + 14}$$

3.4 Naïve Bayes

Example – continued

With Laplace smoothing, we obtain:

$$\begin{aligned}\Pr(S \mid \text{a very close game}) &= \Pr(a \mid S) \times \Pr(\text{very} \mid S) \times \Pr(\text{close} \mid S) \times \Pr(\text{game} \mid S) \times \Pr(S) \\ &= \frac{2+1}{11+14} \times \frac{1+1}{11+14} \times \frac{0+1}{11+14} \times \frac{2+1}{11+14} \times \frac{11}{20} = 2.534 \times 10^{-5} = 0.00002534\end{aligned}$$

$$\begin{aligned}\Pr(NS \mid \text{a very close game}) &= \Pr(a \mid NS) \times \Pr(\text{very} \mid NS) \times \Pr(\text{close} \mid NS) \times \Pr(\text{game} \mid NS) \times \Pr(NS) \\ &= \frac{1+1}{9+14} \times \frac{0+1}{9+14} \times \frac{1+1}{9+14} \times \frac{0+1}{9+14} \times \frac{9}{20} = 6.432 \times 10^{-6} = 0.000006423\end{aligned}$$

Our classifier gives “A very close game” the Sports tag.

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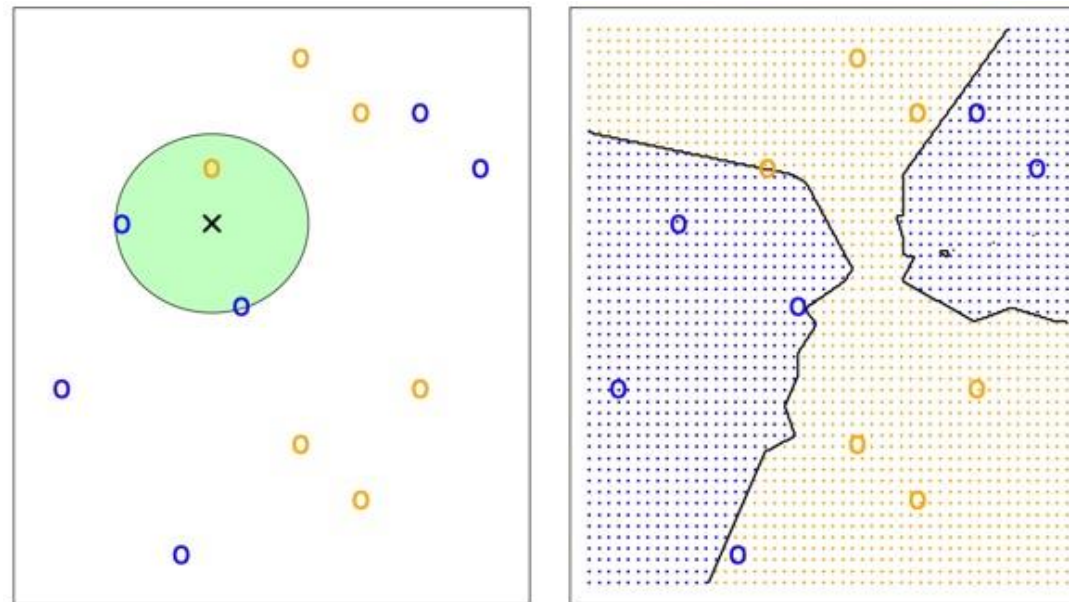
4.5 K-nearest Neighbors

3.5 K-nearest Neighbors

- Given a positive integer K and a test observation t_0 , the KNN classifier first identifies the K points in the training data that are closest to x_0

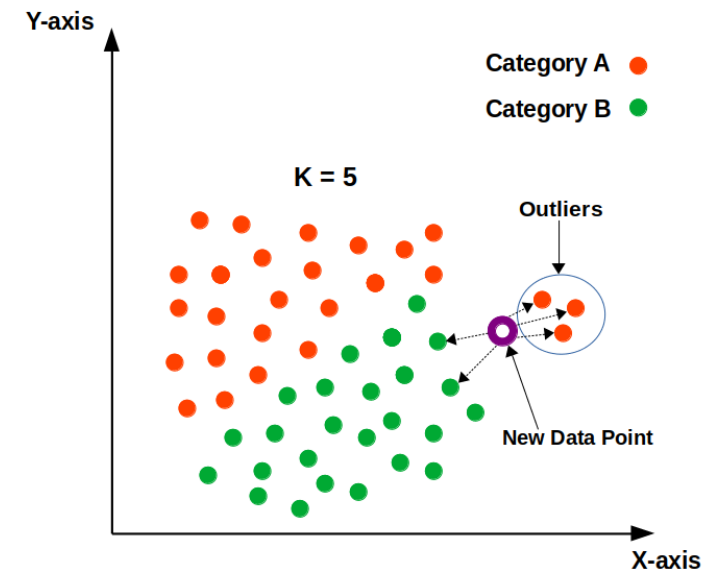
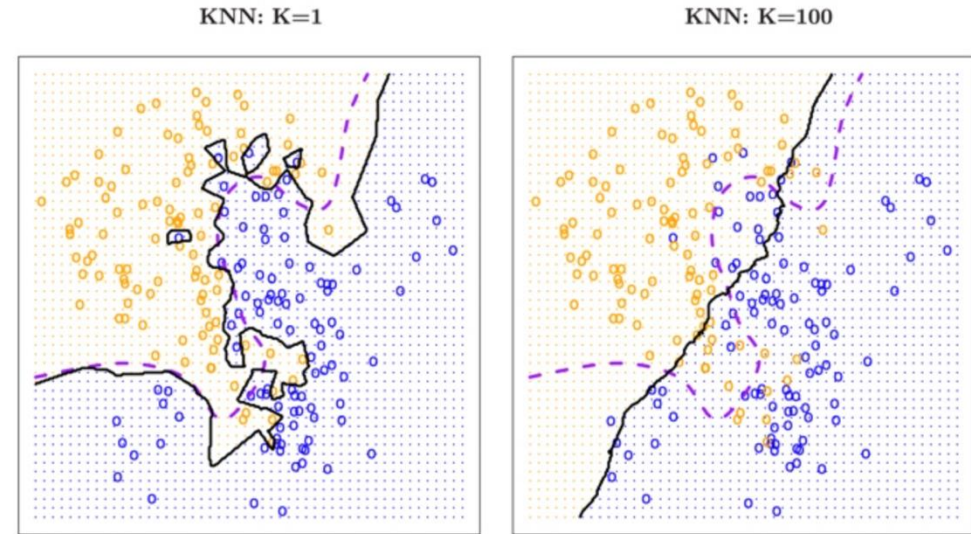
$$\Pr(Y = j \mid X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j)$$

- Then, KNN classifies the test observation x_0 to the class with the largest probability.
- $K = 3$



3.5 K-nearest Neighbors

- With $K = 1$, the decision boundary is overly flexible, while with $K = 100$ it is not sufficiently flexible. The true boundary is shown as a purple dashed line.
- The choice of K has a drastic effect on the KNN classifier obtained.
- Nearest neighbor methods can be lousy when p is large.
- Reason: the **curse of dimensionality**. Nearest neighbors tend to be far away in high dimensions.
- KNN is sensitive to outliers.



Summary

- LDA and logistic regression assume that the log odds of the posterior probabilities are linear in x .
- LDA performs better than logistic regression when the observations at each k^{th} class is normal.
- LDA assumes the features are normally distributed with a common within-class covariance matrix.
- Naive Bayes instead assumes independence of the features.
- K-nearest neighbors (KNN) will be better classifiers when decision boundary is non-linear, n is large, and p is small.
- KNN has low bias but large variance; as such, KNN requires a lot of observations relative to the number of predictors.

Further Reading

- Chapter 4 – James G., Witten D., Hastie T., Tibshirani R., & Taylor J. (2023) *An Introduction to Statistical Learning with Applications in Python*. Springer <https://link.springer.com/book/10.1007/978-3-031-38747-0>
- Chapter 4 - Introduction to Statistical Learning Using R Book Club: <https://r4ds.github.io/bookclub-islr/classification.html>
- Summary of Chapter 4: <https://www.bijenpatel.com/guide/islr/classification/>
- KNN demo - <http://vision.stanford.edu/teaching/cs231n-demos/knn/>

End of Topic

Thank you
Any questions?